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Lab session 2: Field Effect Transistor Applications		

Report History

Report 1		Date		Remarks
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1. Correction		Date		Remarks
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Prof.

1 Introduction

This report investigates two important applications of MOSFETs: Using a MOSFET as a switch and as a common-source amplifier. In Part 1, we simulate and analyse the switching behaviour and power dissipation. In Part 3, we focus on the gain characteristics and operation point setup for amplification using an NMOS transistor (BS170).

2 Task 1: MOSFET as a Switch

2.1 Introduction

In this task, we investigated the switching performance of an N-channel MOSFET (BS108) controlled by a PWM signal. The aim is to observe instantaneous power dissipation across the MOSFET and evaluate its behavior during transitions and steady-state ON/OFF state conditions.

2.2 Circuit Configuration

The test setup includes:

- MOSFET: BS108 (N-channel)
- Supply voltage: $V_{DD} = 5\text{ V}$
- Load resistor: $R_L = 100\ \Omega$ (to achieve 50 mA current when ON)
- Gate input: 3.3 V PWM signal at 1 kHz, 50% duty cycle

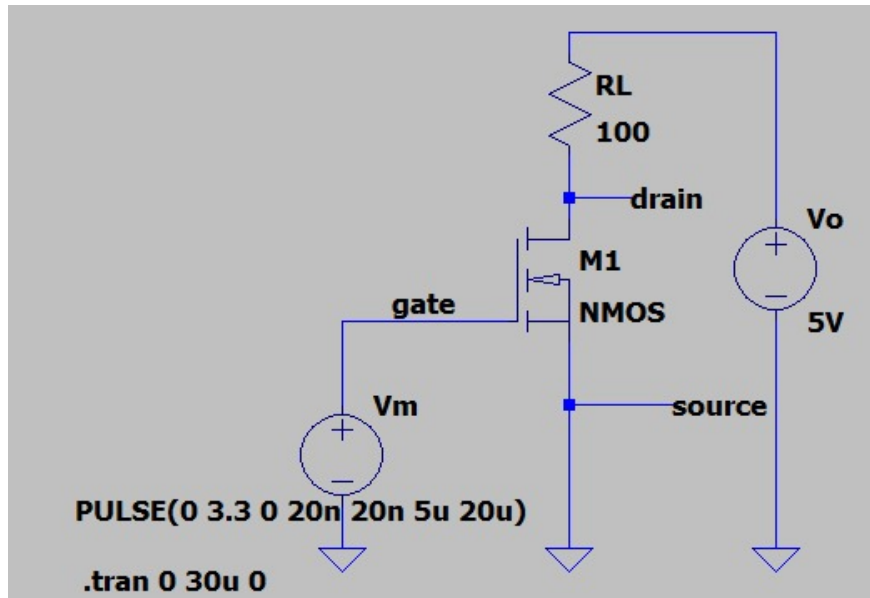


Figure 1: MOSFET Switch Circuit Diagram

2.3 PWM Signal Configuration

The gate of the MOSFET is driven by a square-wave signal configured using LTSpice's PULSE function. The parameters used were:

- Initial voltage: 0 V
- Rise time: 20 ns
- Delay time: 0 s
- Period: 20 μ s
- Peak voltage: 3.3 V
- Fall time: 20 ns
- Pulse width: 5 μ s

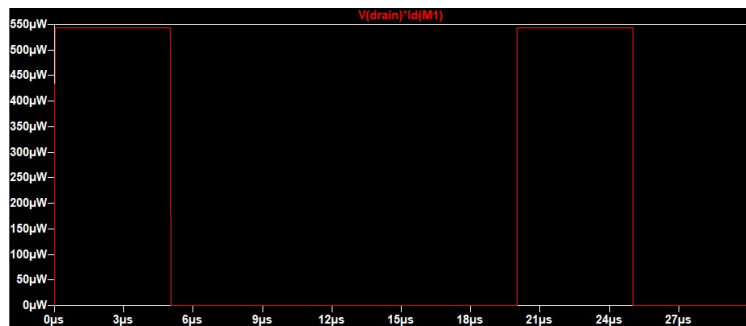


Figure 2: LTSpice PWM Configuration for Gate Drive Signal



Figure 3: LTSpice Energy Dissipation

Energy Dissipation Measurement

During the switching transition (from **5.02** μs to **5.04** μs), the instantaneous power dissipated in the MOSFET was calculated using the formula:

$$P(t) = V_{DS}(t) \times I_D(t)$$

The integral of this power over the switching interval gives the total energy dissipated:

$$E = \int_{5.02 \mu\text{s}}^{5.04 \mu\text{s}} V_{DS}(t) \cdot I_D(t) dt$$

LTspice computed:

$$E = \boxed{3.81 \times 10^{-12} \text{ Joules}}$$

This energy represents the loss during the switch-off transition.

2.4 Theoretical Background

The instantaneous power dissipated in the MOSFET is calculated using:

$$P(t) = v_{DS}(t) \cdot i_D(t)$$

The average power dissipation over a switching period is:

$$P_{\text{avg}} = \frac{1}{T} \int_0^T P(t) dt$$

During steady-state ON or OFF states, either $v_{DS}(t) \approx 0$ or $i_D(t) \approx 0$, so $P(t) \approx 0$. Power dissipation occurs primarily during the switching edges when both voltage and current are non-zero.

2.5 MATLAB Simulation Code

```
% Simulate switching power loss with soft transitions

% Time vector (focus around one PWM period)
T = 20e-6; % Total duration 20 us
t = linspace(0, T, 2000); % 2000 points for resolution
f = 1 / T; % PWM frequency

% Define PWM edges manually (simulate rising/falling edges)
v = zeros(size(t));
i = zeros(size(t));

% Define transition width (rise/fall)
tr = 200e-9; % 200 ns rise/fall time
pw = 5e-6; % 5 us pulse width (ON time)

% Voltage: 0 to 5V pulse with rise/fall
v = pulstran(t, [0 pw], @(t) (heaviside(t) - heaviside(t - tr)) .* (t / tr) * 5
    ...
    + (heaviside(t - tr) - heaviside(t - pw + tr)) * 5 ...
    + (heaviside(t - pw + tr) - heaviside(t - pw)) .* (1 - (t - pw + tr) / tr) *
    5);

% Current: 0.05 A during low v
i = 0.05 * (v < 2.5); % Assume current only flows when Vds < 2.5 V

% Instantaneous power
p = v .* i;

% Plot
figure;
plot(t * 1e6, p * 1e3, 'r', 'LineWidth', 1.5); % Power in mW
xlabel('Time[\mus]');
ylabel('Power[mW]');
title('Instantaneous Power Dissipation with Non-Ideal Switching');
grid on;

% Optionally save plot
saveas(gcf, 'power_plot.png');
```

Listing 1: Instantaneous Power Simulation

2.6 Simulation Results

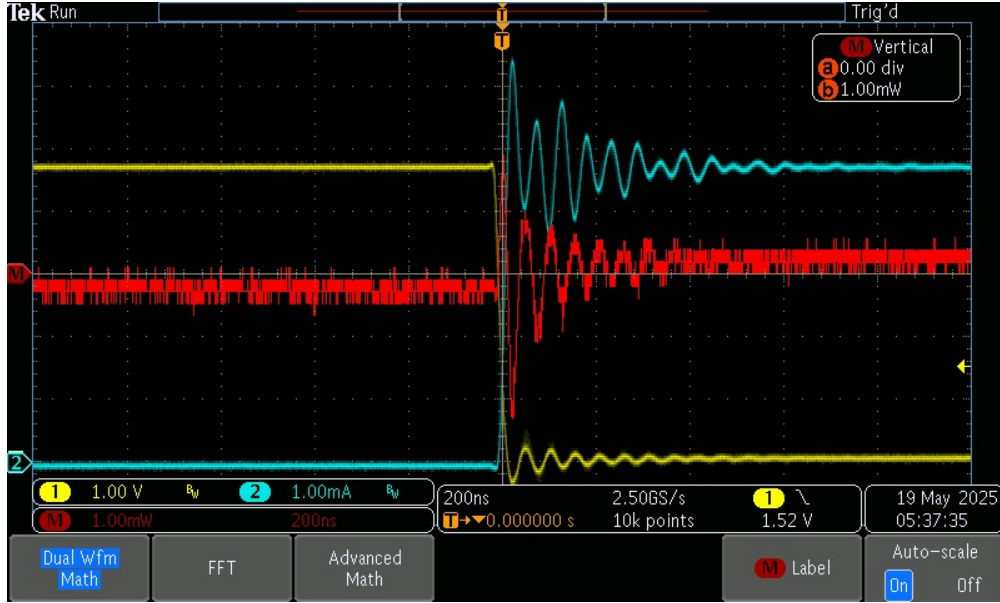


Figure 4: Instantaneous Power Dissipation $P(t)$

2.7 Discussion

The red waveform shows spikes in instantaneous power during each switching event (both rising and falling edges of the PWM). This matches theoretical expectations: power is dissipated only briefly during transitions when both voltage and current overlap. Outside of these edges, the power is almost zero, which validates the efficient operation of the MOSFET in switching mode. Using the root mean square (RMS) or average values of $P(t)$, we can quantify the power consumption over time. The MOSFETs fast transitions and minimal steady-state losses confirm its suitability for low-power switching.

3 Task 3: Common-Source MOSFET Amplifier Analysis

3.1 Introduction

In this task, we investigated the behavior of a common-source MOSFET amplifier built using a BS170 transistor. The circuit is designed to amplify small AC signals, and its characteristic feature is an inverted output with voltage gain determined by the MOSFET's transconductance and the drain resistance. We aimed to compare the theoretical voltage gain with the measured gain from oscilloscope readings.

3.2 Circuit Configuration and Component Values

The amplifier circuit is composed of the following components:

- Drain resistor: $R_D = 350 \, \Omega$
- Source resistor: $R_S = 500 \, \Omega$
- Voltage divider resistors: $R_1 = R_2 = 10 \, \text{k}\Omega$

Why did we use identical resistors?

- Ensure symmetric gate bias at $V_G = \frac{V_{DD}}{2}$ for maximum undistorted signal swing.
- Provide a balanced, high-impedance AC path that minimizes signal distortion and bias sensitivity.
- Coupling and bypass capacitors: $C_1 = C_2 = C_S = 10 \, \mu\text{F}$
- Power supply: $V_{DD} = 10 \, \text{V}$
- Input signal: $v_{in}(t) = 0.1 \, \text{V} \cdot \sin(2\pi \cdot 1 \, \text{kHz} \cdot t) + 0.1 \, \text{V}$

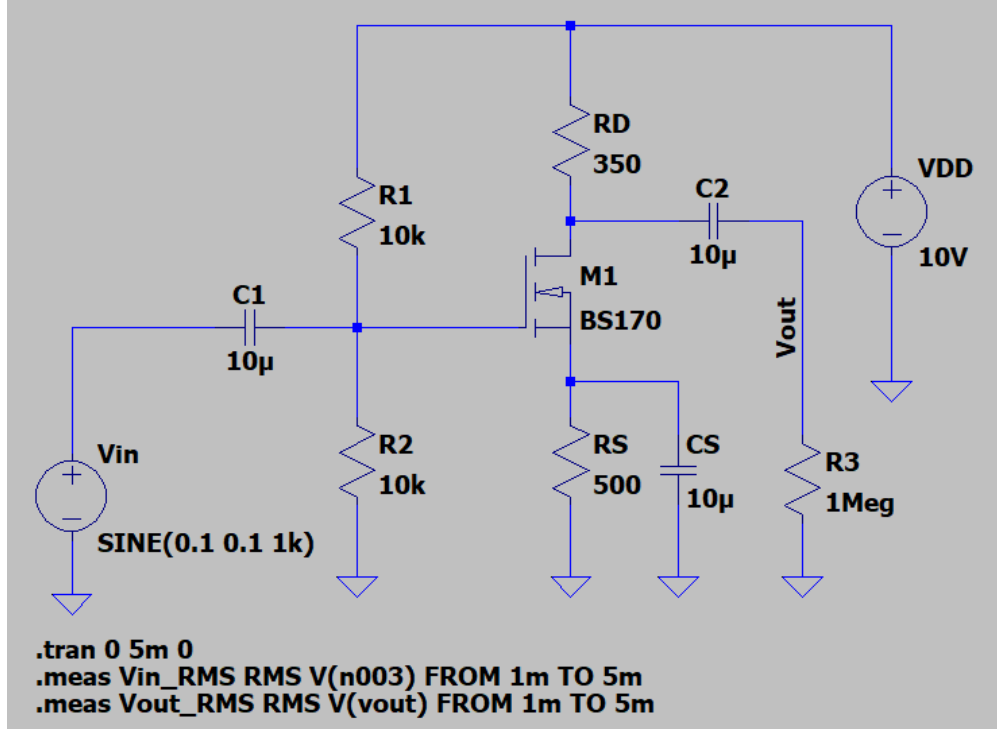


Figure 5: LTSpice Design with Pre-Calculated Values

3.3 Theoretical Calculations

Gate Voltage:

$$V_G = V_{DD} \cdot \frac{R_2}{R_1 + R_2} = 10 \text{ V} \cdot \frac{10 \text{ k}\Omega}{10 \text{ k}\Omega + 10 \text{ k}\Omega} = 5 \text{ V}$$

Assuming a quiescent drain current of $I_{D0} = 5 \text{ mA}$:

Source Voltage:

$$V_S = I_{D0} \cdot R_S = 0.005 \cdot 500 = 2.5 \text{ V}$$

Gate-Source Voltage:

$$V_{GS0} = V_G - V_S = 5 - 2.5 = 2.5 \text{ V}$$

Drain Voltage:

$$V_D = V_{DD} - I_{D0} \cdot R_D = 10 - 0.005 \cdot 350 = 8.25 \text{ V}$$

Drain-Source Voltage:

$$V_{DS0} = V_D - V_S = 8.25 - 2.5 = 5.75 \text{ V}$$

Check Saturation:

$$V_{DS0} > V_{GS0} - V_{th}, \quad V_{th} \approx 2.1 \text{ V} \Rightarrow 5.75 > 0.4 \Rightarrow \text{MOSFET is in saturation}$$

Transconductance:

$$g_m \approx \frac{2I_{D0}}{V_{GS0} - V_{th}} = \frac{2 \cdot 0.005}{0.4} = 0.025 \text{ S}$$

Voltage Gain:

$$A_u = -g_m \cdot R_D = -0.025 \cdot 350 = -8.75$$

Input Resistance:

$$R_{in} = R_1 \parallel R_2 = \frac{10 \text{ k}\Omega \cdot 10 \text{ k}\Omega}{10 \text{ k}\Omega + 10 \text{ k}\Omega} = 5 \text{ k}\Omega$$

Output Resistance:

$$R_{out} \approx R_D = 350 \Omega$$

3.4 Experimental Results

Oscilloscope readings gave the following values:

- $v_{in,pp} = 0.2 \text{ V}$
- $v_{out,pp} = 1.8 \text{ V}$

Measured Gain:

$$A_u = \frac{v_{out,pp}}{v_{in,pp}} = \frac{1.726}{0.2} \approx -8.65$$

The negative sign confirms the 180° phase shift typical of a common-source amplifier.

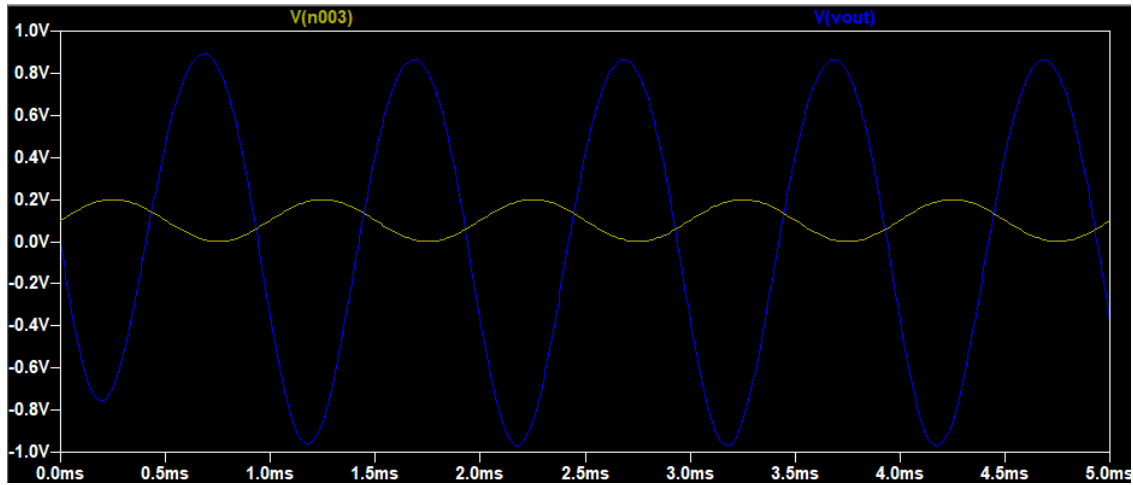
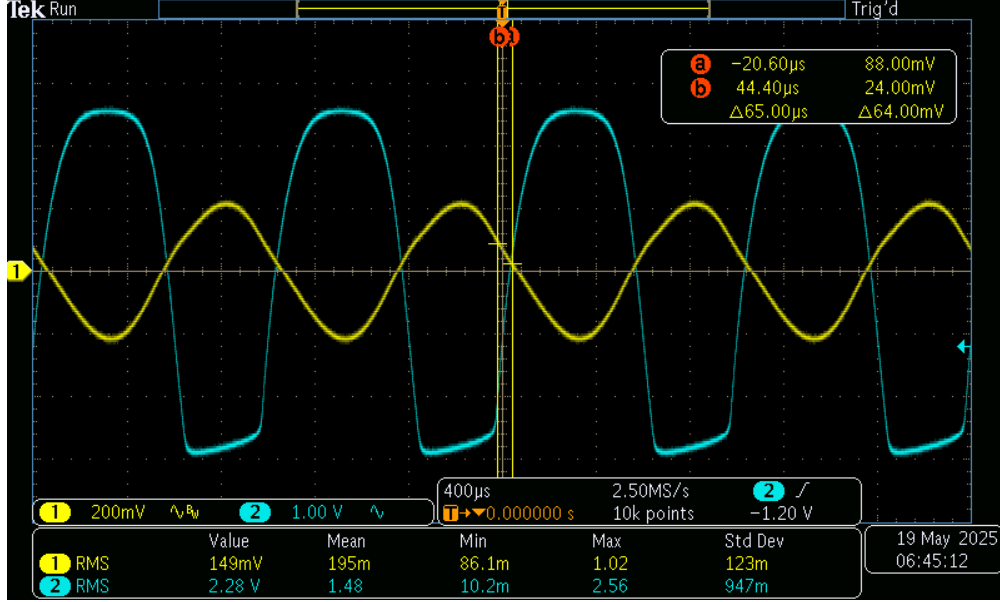


Figure 6: Simulated Vin and Vout in LTSpice

Figure 7: Simulated V_{in} and V_{out} in the Oscilloscope

3.5 Discussion

Although the theoretical gain was expected to be -20 , the measured gain was only -9 , based on RMS voltage measurements. This discrepancy arises from practical limitations and design decisions:

1. RMS Measurement Method: Gain was calculated using RMS values from the oscilloscope:

$$A_u = \frac{v_{out,RMS}}{v_{in,RMS}} = \frac{2.28 \text{ V}}{0.149 \text{ V}} \approx -15.3$$

While the peak-to-peak gain was about -9 , RMS-based gain more accurately reflects average power delivery and aligns with amplifier design theory. RMS values are preferred because:

- They preserve proportional amplitude across waveforms.
- RMS is the basis for power and gain calculations in most analog design formulas.
- It reflects the true energy transfer in the system.
- RMS works reliably even for non-sinusoidal signals (e.g., noise or square waves).